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Studierichting: Informatica
Oefeningenreeks 7A nr. 14.10

$$f(x,y) = \sqrt{x^2+y^2}$$

$$f(3,-4) = \sqrt{3^2+(-4)^2} = \sqrt{9+16} = 5$$

$$\frac{\partial f}{\partial x}(x,y) = \frac{\partial}{\partial x}\sqrt{x^2+y^2} = \frac{2x}{2\sqrt{x^2+y^2}} = \frac{x}{\sqrt{x^2+y^2}}$$

$$\frac{\partial f}{\partial y}(x,y) = \frac{\partial}{\partial y}\sqrt{x^2+y^2} = \frac{2y}{2\sqrt{x^2+y^2}} = \frac{y}{\sqrt{x^2+y^2}}$$

$$\frac{\partial f}{\partial x}(3,-4) = \frac{3}{\sqrt{3^2+(-4)^2}} = \frac{3}{5}$$

$$\frac{\partial f}{\partial y}(3,-4) = \frac{-4}{\sqrt{3^2+(-4)^2}} = \frac{-4}{5}$$

$$z - f(3,-4) = \frac{\partial f}{\partial x}(3,-4)(x-3) + \frac{\partial f}{\partial y}(3,-4)(y-(-4))$$

$$z - 5 = \frac{3}{5}(x-3) - \frac{4}{5}(y+4)$$

$$z = 5 + \frac{3}{5}x - \frac{9}{5} - \frac{4}{5}y - \frac{16}{5}$$

$$z = \frac{3}{5}x - \frac{4}{5}y + \frac{25}{5} - \frac{9}{5} - \frac{16}{5}$$

$$z = \frac{3}{5}x - \frac{4}{5}y$$

# References